

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 01



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Recap of Previous Lecture



Topic

Hierarchical Routing

Topic

Source Routing

Topic

Traffic Shaping



Topics to be Covered



Topic

Error Control



ABOUT ME



Hello, I'm **Abhishek**

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- 12 years of GATE CS teaching experience

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Topic : Error Control



Receiver

(Received data)

1 0 1 1 0 0 1 0

Sender

(Transmitted data)

1 0 1 1 0 0 1 0

→ if "Received data" is not same as "Transmitted data"
then "chance of error"



Topic : Error



Error : Corrupted data [flipped data bits]

Types of error:

1. **Single bit error**
2. **Burst error**



Topic : Single bit error

→ Only one bit in the received data has changed.

Transmitted data = 1 0 1 1 0 0 1 0

Received data = 1 0 1 1 1 0 1 0





Topic : Burst Error



- Multiple bit error
- More than one [two or more] bit in the received data have changed.

Transmitted data = 1 0 1 1 0 0 1 0

Received data = 1 1 1 1 0 1 0 0



How many no. of bits corrupted = 3



Topic : Burst Error



→ Burst Length =
Length from first corrupted bit to the last corrupted bit [inclusive]

Transmitted data = 1 0 1 1 0 0 1 0

Received data = 1 1 1 1 0 1 0 0

↑ ↑ ↑
← B.L. = 6 →

→ In case of burst error,
total number of corrupted data bits is less than equal to Burst Length

In case two bits Error
 $\text{min}^m \text{Burst Length} = 2$ [Consecutive bits may be corrupted]
 $\text{max}^m \text{Burst Length} = \text{Block Size}$

In case of 3 bits Error

$\text{min}^m \text{Burst Length} = 3$ (Consecutive bits are corrupted)

$\text{Max}^m \text{Burst Length} = \text{Block Size}$

#Q. Consider ASCII character "A" is transmitted by transmitter, but ASCII character "D" is received by receiver. Calculate burst length?

$$A = 65$$

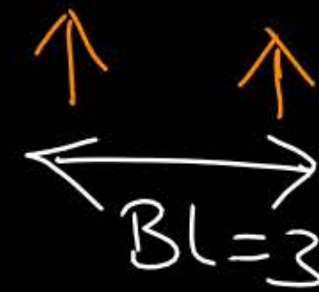
$$D = 68$$

Transmitted Data = 'A'

$$= 65 = 01000001$$

Received Data = 'D'

$$= 68 = 01000100$$



Two bits Error

Burst Length = 3

Ans = 3



Topic : Error Control



→ Based on Redundant bits
[Parity bits or extra bits] check bits

1. Error detection
2. Error detection and correction



Topic : Error detection



- Can only detect error(s)
- Not able to correct
- Retransmission of corrupted data

Two error detection technique :

1. Cyclic Redundancy Check (CRC) (V. Imp.) $\Rightarrow$ Single bit parity (CRC-1)
2. Checksum



Topic : Error detection and correction

- Can detect as well as correct error(s)
- Forward error correction (FEC) $\Rightarrow$ *No Retransmission*

Two error detection and correction technique :

1. 2D Parity $\rightarrow$ (2008)
2. Hamming Code $\rightarrow$ (2021)

V. Imp.
Hamming Distance

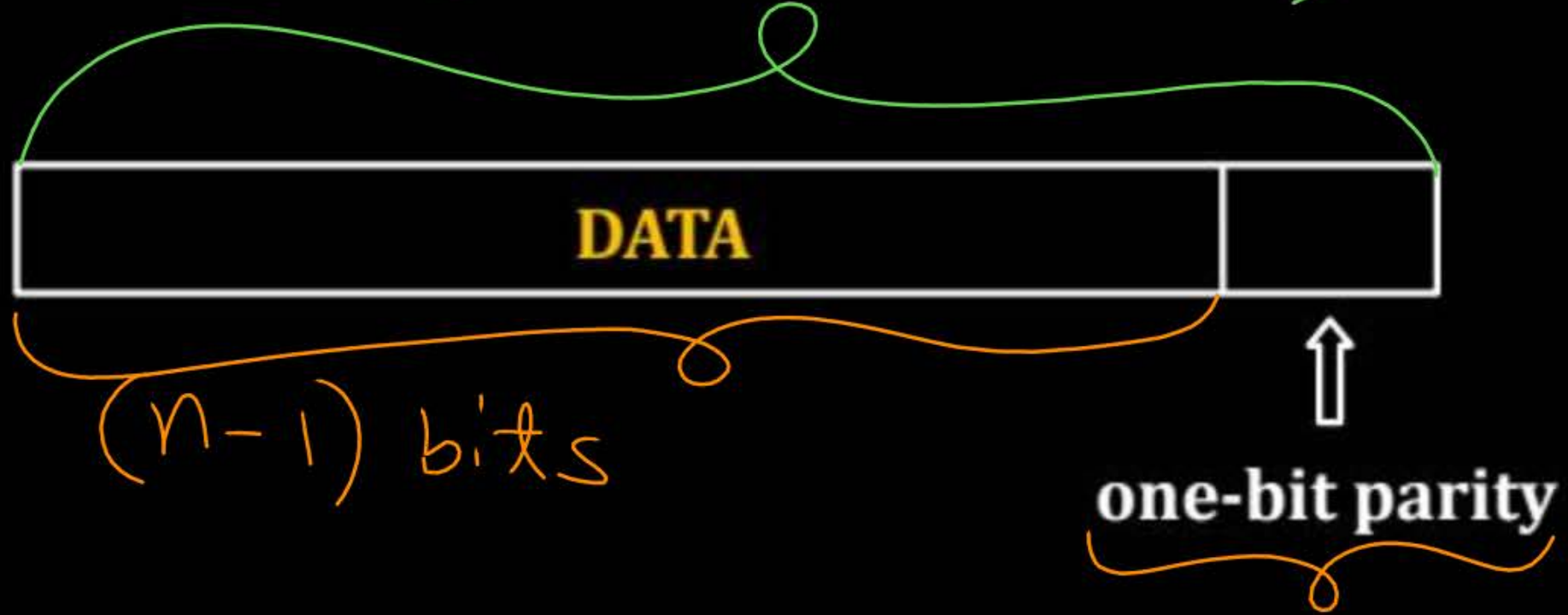


Topic : One-bit parity

[Single bit parity]



Block size (n bits)





Topic : One-bit parity



Transmitter protocol :-

1. Even Parity

if number of one's in the data is even
than transmitter set parity bit "zero"
else
set parity bit "one"

2. Odd Parity

if number of one's in the data is odd
than transmitter set parity bit "zero"
else
set parity bit "one"



Topic : One-bit parity



Suppose “**Even parity**”

DATA = “1 0 1 1 1 0 1”
7 bit

Transmitted Data = 1 0 1 1 1 0 1 1
DATA ↑
parity

Block



Topic : One-bit parity



Receiver protocol :-

1. Even Parity

if receiver find number of one's in the received block is even

then receiver concluded "no error detected"

else

receiver concluded "error detected"

2. Odd Parity

if receiver find number of one's in the received block is odd

then receiver concluded "no error detected"

else

receiver concluded "error detected"



Topic : One-bit parity



Suppose “**Even parity**”

CASE I : No any error

DATA = “1 0 1 1 1 0 1”

Transmitted Data = 1 0 1 1 1 0 1 **1**

Received Data = 1 0 1 1 1 0 1 1

Receiver Concluded : No any error detected, accept the data



Topic : One-bit parity



Suppose “**Even parity**”

CASE II : One-bit error

DATA = “1 0 1 1 1 0 1”

Transmitted Data = 1 0 1 1 1 0 1 **1**

Received Data = 1 0 1 **0** 1 0 1 1

Receiver Concluded : Error detected, reject the data



Topic : One-bit parity



Suppose "**Even parity**"

CASE III : Two-bit error ✓

DATA = "1 0 1 1 1 0 1"

Transmitted Data = 1 0 1 1 1 0 1 **1**

Received Data = 1 0 **0** 1 1 **1** 1 1

0 0 $\Rightarrow$ 1 1

0 1 $\Rightarrow$ 1 0

1 0 $\Rightarrow$ 0 1

1 1 $\Rightarrow$ 0 0

Receiver Concluded : No any error detected, accept the data



Topic : One-bit parity



Suppose “**Even parity**”

CASE IV : Three-bit error ✓

DATA = “1 0 1 1 1 0 1”

Transmitted Data = 1 0 1 1 1 0 1 **1**

Received Data • = 1 0 1 **0 0 1** 1 1

Receiver Concluded : Error detected, reject the data



Topic : One-bit parity ✓



→ Receiver detect “**all single bit error**” ✓

→ In case of burst error,
receiver able to detect “**all odd number of errors**”
↓

#Q. Let suppose, even parity is used in one-bit parity error detection technique. If receiver find total 295 one's in the received block (including parity) then what receiver concluded?

[H.W.]

- (A) No any error detected
- (B) Error detected
- (C) Unable to detect error
- (D) Data insufficient



2 mins Summary



Topic

One-bit parity

Topic

~~Valid vs invalid Codewords~~



THANK - YOU

